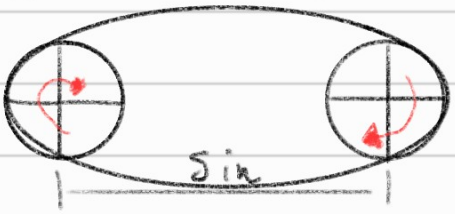
Belt design



Knowns.

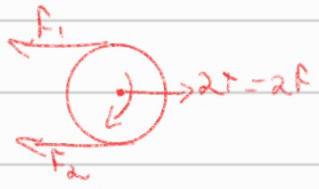
Parallel axes seperated by a distance of 4-6 inches

- Pulley diameter: 3 inches $= d_p$
- Rot. speed: 1000-2000 RPM → Chose: 2000 RPM $= \omega$
- Belt tension: 4-6 lbf → Chose: 6 lbf $= T$
- Pulley material → Steel
- S F (Fos) = 3
- Belt Material → Polyurethane rubber
  - σ_y = 3.25 MPa
  - Strain at 20%

Unknowns

F_1, F_2, d_{belt}

Assuming $\mu = 1$ (No slippage)



$\Sigma F_x = -F_1 - F_2 + 2F = 0$

$2F = F_1 + F_2$

$\frac{F_1}{F_2} = e^{\mu\phi}$ → $\frac{2F - F_2}{F_2} = e^{\mu\phi}$

$\frac{2F}{F_2} = 1 + e^{\mu\phi}$

$F_1 = 2(T) - F_2$

$F_1 = 11.5029$ lbf

$F_2 = \frac{2F}{1+e^{\mu\phi}} = \frac{2(6\ lbf)}{1+e^{(1)(\pi)}}$

$F_2 = 0.4971$ lbf

$\sigma = \frac{S_y}{S.F.} = \frac{F}{A}$

$A_b = \frac{F_1 \cdot S.F.}{S_y}$

$\frac{\pi}{4} d_b^2 = \frac{F_1 \cdot (S.F.)}{S_y}$

$d_b = \sqrt{\frac{4 \cdot F_1 \cdot (S.F.)}{S_y \cdot \pi}}$

$d_b = 0.30542804$ in → Closest industry belt size?

$A_b = \frac{\pi}{4} d_b^2 = 0.073267\ in^2$

$5/16" = 0.3125$ in

Belt part # 3044K707

Belt circumference

$C = 2L + \pi(d_p + d_b)$

$C = 2(6) + \pi(3 + 0.30542804)$

$C = 22.384$ in

Shaft Design

Using 1060 hardened steel → $\sigma_y = 98200\ psi = S_y$

Knowns:	Unknowns
$T = F = 6\ lbf$	τ, d_{min} of shaft
S.F. = 3	
$\varnothing_{Pulley} = 3''$ $= d_p$	
Pulley material: steel	
$F_1 = 11.5024\ lbf$	
$F_2 = 0.4971\ lbf$	
$d_{belt} = 0.305429804''$	
$S_y = 3.25\ Mpa = 471\ psi$ at 20% min	
Rot. speed = 2000 RPM $= \omega$	
$\mu =$	



$T = (F_1 - F_2)\frac{d_p}{2} = (F_1 - F_2)(1.5)$

$T = 16.50874204\ 2\ lbf \cdot in$

$V_{max}(shear_{max}) = 65000\ psi$

$\tau_{max} = \frac{V_{max}}{S.F} = 16250\ psi$ — used website for value

$\tau_{max} = \frac{16T}{\pi d^3}$

$d = \sqrt[3]{16T/\pi\tau_{max}}$

$d = 0.172959632$

$d_s = 0.1730\ in$ — Min diameter of shaft

https://www.makeitfrom.com/material-properties/SAE-AISI-1060-S58C-G10600-Carbon-Steel

Journal Bearings

Knowns:

$T = F = 6$ lbf

S.F. = 3

$\varnothing_{Pulley} = 3'' = d_p$

Pulley material: steel

$F_1 = 11.5029$ lbf

$F_2 = 0.4971$ lbf

$d_{belt} = 0.30542804$ in

$S_y = 3.25$ Mpa $= 471$ Psi at 2% min

Rot. speed $= 2000$ RPM $= \omega$

$\nu = 1$

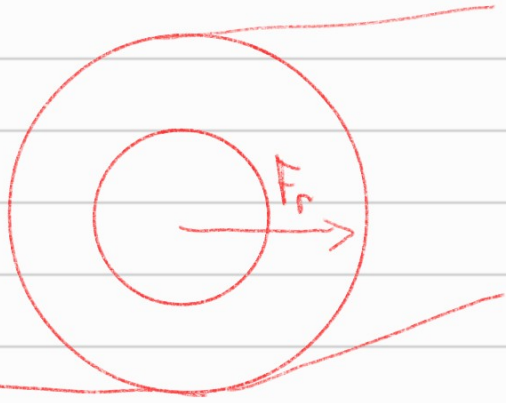
$d_{shaft} = 0.1730$ in

$\tau_{shaft} = 16250$ lb·in

$\sigma_{shaft} = 98200$ Psi

Unknowns

f_t, f_r, P, V



$$f_t = \frac{2T}{d_s} = 64.38054 \text{ lb}$$

$$f_r = f_t \tan\alpha = F_t \tan(20^\circ) = 25.2525 \text{ lb}$$

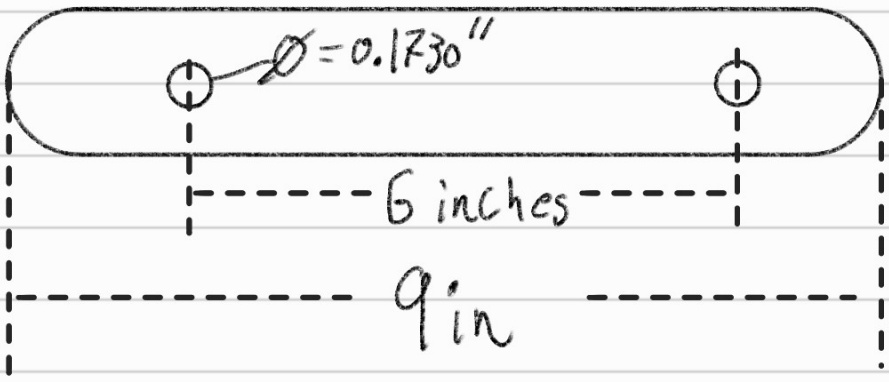
$$P = \frac{F}{A} = \frac{f_r}{L \cdot d} = \frac{F_r}{3\text{in} \cdot (d_{shaft})} = 48.6674632 \text{ Psi}$$

$$V = \omega \pi d_{shaft} = 90.5612061 \text{ ft/min}$$

$$PV = 4,407.3842 \text{ psi} \cdot \text{ft/min} = PV$$

Bearing Material	Load Capacity (psi)	Maximum Temperature (°F)	Surface Speed, V_{max} (max. fpm)	PV Limit P = psi load V = surface ft/min
~~Acetal~~	~~1000~~	~~180~~	~~1000~~	~~3000~~
Graphite (dry)	600	750	2500	15,000
Graphite (lubricated)	600	750	2500	150,000
~~Nylon, Polycarbonate~~	~~1000~~	~~200~~	~~1000~~	~~3000~~
Nylon composite	…	400	…	16,000
Phenolics	6000	200	2500	15,000
Porous bronze	4500	160	1500	50,000
Porous iron	8000	160	800	50,000
Porous metals	4000–8000	150	1500	50,000
~~Virgin Teflon (TFE)~~	~~500~~	~~500~~	~~50~~	~~1000~~
Reinforced Teflon	2500	500	1000	10,000–15,000
TFE fabric	60,000	500	150	25,000
Rubber	50	150	4000	15,000
Maple & Lignum Vitae	2000	150	2000	15,000

Frame Design

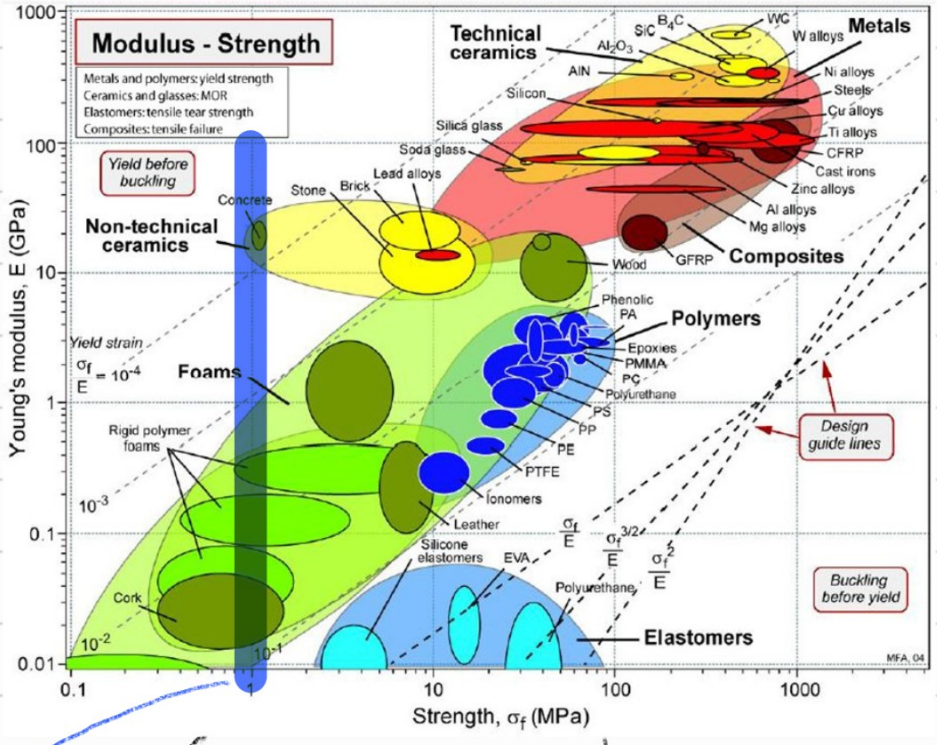


48.6674632 psi = 0.33555 MPa

S.F. x Pressure = 3 x 0.33555 MPa = 1.00665 MPa

Using Ashby plots to simplify the material selection process.

Stronger



lighter

located around the rigid foams, should withstand the pressure the bearings will encounter.